NB2 Content Clinic

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60 seconds per question

Answers & explanations will be on each following slide

Slides will be shared after the session

A 54-year-old man comes to the physician because of a 2-week history of muscle weakness. He was diagnosed with bronchial carcinoma 6 months ago. His temperature is 37.5oC (99.5oF), pulse is 71/min, blood pressure is 130/85 mm Hg. An EMG shows a waxing pattern. Which of the following is the most likely diagnosis?

A. Guillain Barre syndrome

B. Lambert-Eaton syndrome

C. Muscular dystrophy

D. Myotonia congenita

E. Myasthenia gravis

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A 40-year-old man comes to the physician due to uncontrollable arm movements. About a week ago he started having episodes where he was violently flinging his right arm. His left arm is not experiencing these episodes. MRI of the brain shows a lesion on the left subthalamic nucleus. What disorder does this patient most likely have?

A. Huntington's Disease

B. Myasthenia Gravis

C. Hemiballismus

D. Lambert Eaton

E. Parkinson Disease

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A 4-year-old boy is brought to the physician by his mother due to progressive weakness. He frequently falls down and has to use a special maneuver to stand up. He has hypertrophic muscles in his calf. Blood test shows high levels of CK. What kind of mutation does this boy most likely have?

A. Autosomal Dominant

B. Autosomal Recessive

C. X-linked Dominant

D. X-lined Recessive

E. Mitochondrial

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A 63-year-old man comes to the physician because of a 2-month history of tremors and constipation. Physical examination shows decreased facial expression, rigidity, and postural instability. Neurologic examination shows a slow, shuffling gait and tremor of the right hand at rest. The most likely cause of his condition is degeneration of which of the following tissues:

A. Striatum

B. Thalamus

C. Subthalamic nucleus

D. Globus pallidus

E. Substantia nigra

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A 52-year-old man comes to the physician because of a 2-week history of forgetfulness. He admits that he has been drinking 4 to 5 glasses of scotch every night for the past year. Neurological examination shows truncal ataxia and horizontal nystagmus. Psychological examination shows retrograde and anterograde amnesia. MRI of the head shows atrophy of the mammillary bodies. Which of the following vitamin deficiencies is the most likely cause of these symptoms?

A. Vitamin B1

B. Vitamin B2

C. Vitamin B6

D. Vitamin B9

E. Vitamin B12

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A 68-year-old man comes to the physician because of generalized muscle weakness and fatigue. He is a cigarette smoker with a 60-pack/year history. His pulse is 70/min, blood pressure is 120/80 mm Hg, temperature is 37°C (97°F), and respirations are 15/min. Physical examination shows that his reflexes are initially absent, although they are recovered after a brief period of exercise. Which of the following pharmacotherapies is most suitable for this patient?

A. 4-aminopyridine

B. Steptomycin

C. Atropine

D. Botulinum toxin

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A 34-year-old man is brought to the emergency department because of weakness and fatigue that began 2 weeks ago following a gastrointestinal bacterial infection. The weakness began in his feet and has been slowly ascending towards his trunk. Neurological examination shows lower motor neuron signs in the lower limbs. Nerve conduction studies show slowed conduction velocities of peripheral motor nerves. Which of the following is the most likely diagnosis?

A. Lambert-Eaton Syndrome

B. Myasthenia Gravis

C. Guillain-Barre Syndrome

D. Ducchenne Muscular Dystrophy

E. Poliomyelitis

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A 27-year-old woman comes to the physician because of a 2-month history of hearing loss, ringing, dizziness, and facial weakness. Weber's Test shows lateralization to the right and Rinne’s Test shows air conduction is greater than bone conduction bilaterally. A head CT shows a mass in the left internal acoustic meatus. Which of the following describes the pathogenesis of this patient’s disease?

A. Combined conductive and sensorineural hearing loss of left ear

B. Conductive hearing loss of left ear

C. Conductive hearing loss of right ear

D. Sensorineural hearing loss of left ear

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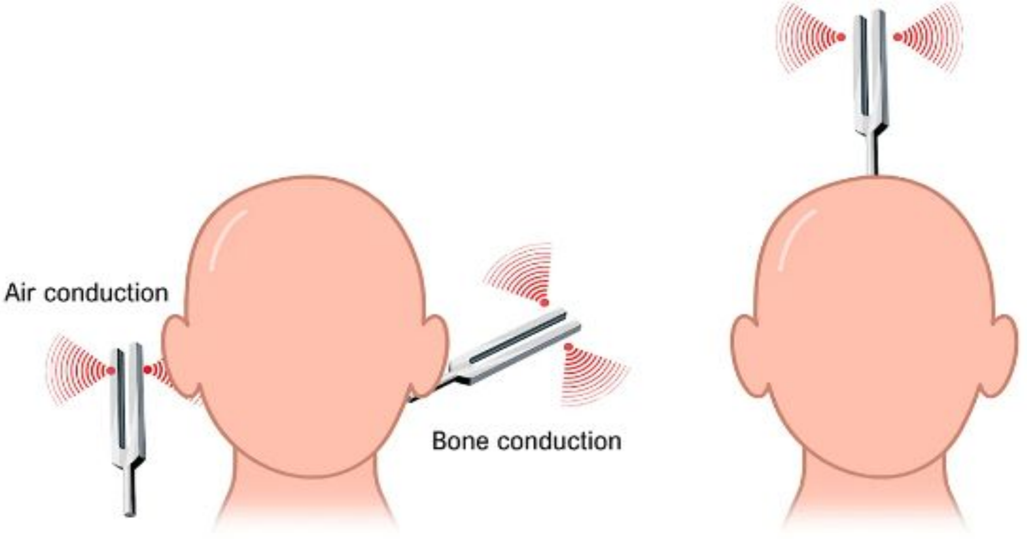

Rinne test | **Weber test**

	Rinne test (Conduction)	Weber test (Localization)
Normal	Air > Bone in both ears	Midline
Conductive hearing loss	Bone > Air in affected ear	Lateralizes to affected ear
Sensorineural hearing loss	Air > Bone in both ears	Lateralizes to normal ear

Illustration by ***Ahmed Afifi***

A 33-year-old man is brought to the emergency department after being extracted from a vehicle involved in a car crash. His pulse is 50/min, blood pressure is 110/75 mm Hg, temperature 37°C (97°F), and respirations are 12/min. Imaging studies show a spinal cord hemisection at the level of T4. Which of the following patterns of sensory loss would be present on neurological exam?

A. Ipsilateral loss of touch, vibration, and proprioception T4 and below; contralateral loss of pain & temperature T6 and below

B. Contralateral loss of touch, vibration, and proprioception T4 and below; contralateral loss of pain & temperature T6 and below

C. Ipsilateral loss of touch, vibration, and proprioception T4 and below; ipsilateral loss of pain & temperature T6 and below

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A 19-year-old-man is brought to the emergency department after falling from the third story of a building. His pulse is 50/min, blood pressure is 110/75 mm Hg, temperature 37°C (97°F), and respirations are 12/min. Imaging studies show extensive bilateral damage to T2-T3 spinal segments. Which of the following reflex patters will be apparent 4 weeks after the injury?

A. Upper Limb 0; Lower Limb 0

B. Upper Limb 2+; Lower Limb 4+

C. Upper Limb 2+; Lower Limb 2+

D. Upper Limb 4+; Lower Limb 4+

E. Upper Limb 4+; Lower Limb 0

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E. Upper Limb 4+; Lower Limb 0

A 27-year-old man comes to the emergency department with second degree burns of his left hand. The patient says he had recently injured his right hand as well while cooking because he could not sense the temperature. He was seen in the emergency department 2-months prior after being involved in a motor vehicle accident. Neurological examination shows sensation to pain and temperature is decreased in the upper extremities and over the shoulders bilaterally. Sensation to light touch and vibration remains intact over the upper extremities bilaterally. Which of the following is the most likely diagnosis?

A. Guillain-Barre

B. Multiple Sclerosis

C. Syringomyelia

D. Brown Sequard Syndrome

E. Transection of the spinal cord

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A 70 year old man is brought to the clinic with rigidity, his wife reports that he walks in short shuffling movements and is hunched over. He prescribed first line treatment, what is the rationale behind this treatment?

A. L-DOPA/Carbidopa: is able to enter the CNS and act on the direct pathway to facilitate movement while preventing peripheral breakdown

B. Antiviral therapy is used to increase the activity of the indirect pathway to inhibit unwanted movement.

C. Dopamine metabolism inhibitors, to prevent the degradation of dopamine in the peripheral nervous system

D. Anticholinergics, which cause a decrease in the direct pathway

E. D2R agonist

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A 23-year-old woman comes to the physician because of severe sunburn on her shoulder and back. Her pulse is 105/min, blood pressure is 140/85 mm Hg and respirations are 15/min. Physical examination shows that light touch of the affected area elicits pain. Which of the following clinical terms most appropriately applies to her physical examination findings?

A. Hyperalgesia

B. Hyperthermia

C. Hypoalgesia

D. Allodynia

E. Paraesthesia

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A 34-year-old man is brought to the emergency department because a motor vehicle accident that resulted from reckless driving. The patient suffers from profound unilateral motor losses with additional somatosensory losses. Digital imaging shows a lesion of the left half of the spinal cord at the T5 level. Which of the following sensory losses will most likely be observed in this patient?

A. Contralateral loss of touch, vibration, proprioception at T5

B. Contralateral loss of touch, vibration, proprioception below T5

C. Contralateral loss of pain and temperature at T5

D. Ipsilateral loss of pain and temperature below T5

E. Ipsilateral loss of pain and temperature at T5

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A 32-year-old man comes to the physician for a follow-up examination. He has a progressive 2-year history of motor impairment in his upper extremities. His pulse is 70/min, blood pressure is 120/80 mm Hg, temperature is 37°C (97°F), and respirations are 12/min. Imaging shows a fluid-filled cavity within the central region of the cervical spinal cord between C2-C7 and a diagnosis of syringomyelia is made. Which of the anterolateral deficits will be seen in this patient?

A. Bilateral loss of pain and temperature at C2-C7

B. Bilateral loss of pain and sensation at C3-C6

C. Bilateral loss of pain and temperature at C4-C5

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E. Bilateral loss of pain and temperature From C5 and below

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An 83-year-old man comes to the emergency department because of a 1-hour history of weakness in his face and left arm and leg. Physical examination shows hypertonia of the left upper and lower extremities, but normotonia of the right upper and lower extremities. Drooping of the lower face is noted, but he is able to raise his eyebrows bilaterally. Which of the following additional findings is likely to be found in this patient on reflex testing?

A. Left ankle reflex 1+

B. Left biceps reflex 2+

C. Left biceps reflex 0

D. Left biceps reflex 4+

E. Left ankle jerk 0

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